JUNQI LU

GitHub — Google Scholar — Personal Website
Date of Birth: August 24, 2002 — Email: lu.junqi.001@student.uni.lu
Luxembourg

EDUCATION

University of Luxembourg — Luxembourg

- **Master of Mathematics** — Sep. 2026 – Jun. 2028 (Expected)

Beijing Institute of Technology — Beijing, China

- **B.E. in Computer Science** (Second Bachelor's Degree) — Sep. 2024 – Jun. 2026
- **B.S. in Mathematics and Applied Mathematics** — Oct. 2020 – Jun. 2024

PUBLICATIONS

[1] **Junqi Lu**, Bosen Liu, Cuicui Pei, Qingan Qiu*, and Li Yang*. Learning to optimize termination decisions under hybrid uncertainty of system lifetime and task duration. *Computers & Industrial Engineering*, 206, 111208, 2025. DOI: 10.1016/j.cie.2025.111208 (**Published, IF=6.5, JCR Q1**)

[2] **Junqi Lu**, Ruixiang Sun, Xin Li*, et al. Prevailing Bisimulation Metric Learning Is Biased: Implicit Regularization and Its Remedy. (**Under Review**)

[3] **Junqi Lu**, Qingan Qiu*, Rongchi Sun, and Yike Zhang. Simplicity Wins: How Myopic Decisions Enable Safe Learning in Mission-Critical Systems. (**Under Review**)

[4] Rongchi Sun, Yike Zhang, **Junqi Lu**, and Qingan Qiu*. Dynamic Switching Strategy for Standby Systems with Switching Time Redundancy. (**Under Review**)

RESEARCH EXPERIENCE

Deep Reinforcement Learning & State Representation — Jul. 2025 – Present

Research Assistant | Advisor: Prof. Xin Li | 1029 DRL Lab | Beijing Institute of Technology

- **Theory & Methodology**: Identified the metric learning "Variance Trap" via **SDEs** and proposed **BSPO** (*Bisimulation Saddle-Point Optimization*), a primal-dual framework utilizing an auxiliary network to decouple structural error from transition noise.
- **Evaluation**: Evaluated BSPO on **DeepMind Control Suite** under stochastic dynamics and visual distractions, and on **Atari**, comparing against MICo and SimSR baselines.
- **Outcome**: First-author paper currently under review.

Reliability Engineering & Stochastic Control — Jun. 2023 – Present

Research Assistant | Advisor: Prof. Qingan Qiu | Beijing Institute of Technology

- **Adaptive Control Framework**: Formulated condition-based task termination for multi-state systems under dual uncertainty as an infinite-horizon **POMDP**, integrating real-time Bayesian inference with right-censored data.
- **Theory & Algorithm**: Proved suboptimality of static policies and derived a closed-form $\mathcal{O}(1)$ **Myopic Heuristic**. This computation-free strategy natively embeds an "Exploration Buffer" and acts as a dynamic risk filter, asymptotically outperforming the complete-information Oracle.
- **Publications**: 1 first-author paper in **CAIE** (Q1, IF=6.5); 1 first-author under review at **POM**; 1 co-authored under review at **RESS**.

SELECTED PROJECTS

Nitrogen Cycling Dynamics Model (MCM-ICM 2025 Finalist) [GitHub: 100+ Stars]

- **Modeling**: Developed a dynamic ecosystem model using modified **Lotka-Volterra equations** to simulate nitrogen flow across forest and agricultural food webs.
- **Open Science**: Released a fully reproducible solution repository with comprehensive documentation and tutorials. Garnered **100+ GitHub Stars**, widely recognized by the modeling community for its code quality and educational value.
- **Award**: Selected as a **Finalist Winner (Top 2%)** out of 27,456 teams globally.

Formal Verification of a Mahjong Winning-Hand Checker [Demo] [Code]
AI4Math Summer School, Zhejiang University Summer 2026

- **Independent Project**: Conceived and completed the project individually, from problem selection and specification design to implementation, formal proofs, and the final summer-school presentation.
- **Formal Modeling**: Encoded three-suit Mahjong in **Lean 4**, separating declarative winning-hand specifications from an executable checker for standard hands and seven pairs, including declared melds and kongs.
- **Correctness Proofs**: Proved the executable checker sound and complete with respect to the declarative specification for every hand state: `isWinning state = true` ↔ `WinningSpec state`, extending validation beyond individual test cases.

HONORS & AWARDS

- **Meritorious Winner**, Mathematical Contest in Modeling (MCM) 2026, 2023
- **Outstanding Student Award** (Top 10% university-wide), BIT Nov. 2025
- **Academic Excellence Scholarship** (Rank 1st in CS Program), BIT Oct. 2025
- **Finalist (Top 2%)**, Interdisciplinary Contest in Modeling (ICM) Feb. 2025
- **University "Excellent Thesis" Prize**, BIT Jun. 2024
- **Second Prize**, China Undergraduate Mathematical Contest in Modeling (CUMCM) Nov. 2022

SKILLS

Programming	**Python** (Expert in PyTorch, NumPy, Pandas), **C++** (Linux System Dev), SQL, MATLAB, Bash
Languages	Chinese (Native), English (B2), German (A1)
AI Workflow	Proficient in AI-assisted coding (VS Code + Copilot/Cursor) for efficient development.
Tools	Git, Docker, LaTeX, Makefiles, Lean 4 (Formal Proof Verification)
Outreach	**Tech Blogger** (CNBlogs) with 20K+ views (20+ articles on DSA & LeetCode).

INTERESTS

- **Marathon Running:** Committed long-distance runner with an annual mileage exceeding **1,000 km** for three consecutive years. Half-Marathon Personal Best: **1:41:34**.
- **Music:** Lead guitarist and bassist for university rock bands. Passionate about classic rock arrangement and performance; active participant in campus music festivals.